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MTH 103 — Past Questions & Materials
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Course: MTH 103
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 4
Levels: 100

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Paper 1 of 4 — MTH 103 2019 2020 PastQuestion

Level 100 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN

COLLEGE OF SCIENCE

DEPARTMENT OF MATHEMATICS

SECOND SEMESTER EXAMINATION 2019/2020

Course code & title: MTH 123 -Vectors & Dynamics. Duration: 2HRS

15MINS

PAST QUESTIONS

1. Given that $\vec{A} = 2\vec{y} + \vec{A} + 2\vec{a}$. Find the square of $|\vec{y}| + 1$. (a) 3 (b) 4 (c) 9 (d) 10

:2 1

2. Find the unit vector for the vector $\vec{B} = 3\vec{y} + 2\vec{A} + 5\vec{a}$. (a) $\frac{1}{\sqrt{34}}(3\vec{y} + 2\vec{A} + 5\vec{a})$ (b) $\frac{1}{\sqrt{34}}(3\vec{y} + 4\vec{A} + 5\vec{a})$

:2

:2 1

(c) $\frac{1}{\sqrt{34}}(3\vec{y} + 4\vec{A} + 5\vec{a})$ (d) $\frac{1}{\sqrt{34}}(3\vec{y} + 4\vec{A} + 5\vec{a})$

5 5:2

26

3. Given $\vec{A} = 4\vec{y} + \vec{A} + 3\vec{a}$ & $\vec{B} = 2\vec{j} + 8\vec{k}$. Obtain $\vec{y} \cdot \vec{B}$. (a) 30 (b) 26 (c) 22.8 (d) :520

:2

4. Given that $\vec{A} = 3\vec{y} + 2\vec{A} + 5\vec{a}$ & $\vec{B} = \vec{i} + 2\vec{j} + 2\vec{k}$. calculate the cosine angle between $\vec{A}$ & $\vec{B}$. (a) 6
1

(b) 5 (c) 15:2 (d) 15:2

5. Given that $\vec{A}$ is a function for the vectors $\vec{A}$ and $\vec{B}$. Thus, if $\vec{A}(u) \cdot \vec{B}(u) = 0$. Then, ...

(a) Parallel (b) unit vector (c) perpendicular (d) scalar triple product

6. If $\vec{A} \cdot \vec{B} = 0$ and $\vec{A}$ and $\vec{B}$ are non-zero vectors, then $\vec{A}$ and $\vec{B}$ are said to be...

(a) Parallel (b) unit vector (c) perpendicular (d) scalar triple product

7. Which of the following represent a continuous vector

$\vec{y}(\vec{A} + \vec{A}) + 2\vec{y}(\vec{A})$ $\vec{y}(\vec{A} + \vec{A}) + 2\vec{y}(\vec{A})$

(a) $\lim_{\vec{A} \rightarrow \vec{A}} \vec{A}$

(b) $\lim_{\vec{A} \rightarrow \vec{A}} \vec{y}(\vec{A}) = \vec{y}(\vec{A}_0)$ (d) $\lim_{\vec{A} \rightarrow \vec{A}} \vec{y}(\vec{A}) = \vec{y}(\vec{A}_0)$

$\vec{A} \rightarrow \vec{A} \vec{A} \rightarrow \vec{A} \vec{A} \rightarrow \vec{A} \vec{A} \rightarrow \vec{A}_0 \vec{A} \rightarrow \vec{A}$

$\vec{y} \vec{y}$

8. If $\vec{y} \cdot (\vec{p} \times \vec{y}) = (\vec{y} \times \vec{y}) \cdot \vec{p} = 0$. Then the differential of $\vec{A}$ is given as (a) $\vec{y} \cdot \vec{A} + \vec{y} \cdot \vec{p} + \vec{y} \cdot \vec{y}$

$\vec{y} \cdot \vec{p} + \vec{y} \cdot \vec{y} + \vec{y} \cdot \vec{p}$ (b) $\vec{y} \cdot \vec{p} + \vec{y} \cdot \vec{y} + \vec{y} \cdot \vec{p}$ (c) $\vec{y} \cdot \vec{p} + \vec{y} \cdot \vec{y} + \vec{y} \cdot \vec{p}$ (d) $\vec{y} \cdot \vec{p} + \vec{y} \cdot \vec{y} + \vec{y} \cdot \vec{p}$

$\vec{y} \cdot \vec{p} + \vec{y} \cdot \vec{y} + \vec{y} \cdot \vec{p}$ (b) $\vec{y} \cdot \vec{p} + \vec{y} \cdot \vec{y} + \vec{y} \cdot \vec{p}$ (c) $\vec{y} \cdot \vec{p} + \vec{y} \cdot \vec{y} + \vec{y} \cdot \vec{p}$ (d) $\vec{y} \cdot \vec{p} + \vec{y} \cdot \vec{y} + \vec{y} \cdot \vec{p}$

$\vec{y} \cdot \vec{p} + \vec{y} \cdot \vec{y} + \vec{y} \cdot \vec{p}$ (b) $\vec{y} \cdot \vec{p} + \vec{y} \cdot \vec{y} + \vec{y} \cdot \vec{p}$ (c) $\vec{y} \cdot \vec{p} + \vec{y} \cdot \vec{y} + \vec{y} \cdot \vec{p}$ (d) $\vec{y} \cdot \vec{p} + \vec{y} \cdot \vec{y} + \vec{y} \cdot \vec{p}$

$\vec{y} \cdot \vec{p} + \vec{y} \cdot \vec{y} + \vec{y} \cdot \vec{p}$ (b) $\vec{y} \cdot \vec{p} + \vec{y} \cdot \vec{y} + \vec{y} \cdot \vec{p}$ (c) $\vec{y} \cdot \vec{p} + \vec{y} \cdot \vec{y} + \vec{y} \cdot \vec{p}$ (d) $\vec{y} \cdot \vec{p} + \vec{y} \cdot \vec{y} + \vec{y} \cdot \vec{p}$

9. Express $\vec{y} \cdot (\vec{y} \times \vec{p})$ as a product. (a) $\vec{y} \cdot \vec{p} + \vec{y} \cdot \vec{y} + \vec{y} \cdot \vec{p}$ (b) $\vec{y} \cdot \vec{p} + \vec{y} \cdot \vec{y} + \vec{y} \cdot \vec{p}$ (c) $\vec{y} \cdot \vec{p} + \vec{y} \cdot \vec{y} + \vec{y} \cdot \vec{p}$ (d) $\vec{y} \cdot \vec{p} + \vec{y} \cdot \vec{y} + \vec{y} \cdot \vec{p}$

$\vec{y} \cdot \vec{p} + \vec{y} \cdot \vec{y} + \vec{y} \cdot \vec{p}$

$$\dot{y} * \frac{d}{dt} \dot{y} + \frac{d}{dt} \dot{y} * p$$

$$2 \ 2 \ 2 \ 2 \ 2$$

10. Given that $(\dot{y}, p, \ddot{y}) = \dot{y} \ \dot{p} \ \ddot{y}$. Find ". (a) $\dot{p} \ddot{y} + \dot{y} \ddot{A} + 2\dot{y} \dot{p} \ddot{a}$ (b) $2\dot{y} \dot{p} \ddot{y} + \dot{y} \ddot{A} + \dot{y} \dot{p} \ddot{a}$ (c)

$$2\dot{p} \ddot{y} + \dot{y} \ 2 \ \ddot{A} + \dot{y} \ 2 \ \dot{p} \ddot{a} \text{ (d) } 2\dot{y} \dot{p} \ddot{y} + \dot{y} \ 2 \ \ddot{A} + \dot{y} \ 2 \ \dot{p} \ddot{a}$$

11. If $' = \dot{y} \ 2 \ \dot{p} \ \ddot{y} \ \ddot{A} \ \dot{y} = 2\dot{y} \ddot{y} + \dot{p} \ddot{A} \ 2 \ \dot{y} \ \dot{p} \ 2 \ \ddot{a}$. Find $\ddot{y} \ddot{A} \ddot{y} (\dot{y}) \ \ddot{y} \ \ddot{a} / \ddot{a} \ \dot{C} \dot{y} \dot{A} \ddot{a}$ (1, 21, 1). (a) $25\ddot{y} \ 2 \ \ddot{A}$ (b)

$$5\ddot{y} + \ddot{A} \text{ (c) } 2\dot{y} \ 2 \ \dot{p} \ \ddot{y} \ 2 \ \ddot{y} + \dot{y} \ 2 \ \dot{p} \ 2 \ \ddot{A} \ 2 \ \dot{y} \ 3 \ \dot{p} \ 3 \ \ddot{a} \text{ (d) } (23\dot{y} \ 2 \ \dot{p} \ 2 \ \ddot{y} \ 2 \ 2\dot{y} \ 2 \ \dot{p} \ 2 \ \ddot{y}) \ddot{y} \ 2 \ (3\dot{y} \ 2 \ \dot{p} \ 3 \ \ddot{y} \ 2 \ 4\dot{y} \ 3 \ \dot{p} \ \ddot{y}) \ddot{A} +$$

$$2 \ 2 \ 3$$

$$(2\dot{y} \dot{p} \ \ddot{y} \ 2 \ 2\dot{y} \ \ddot{y}) \ddot{a} \ 2$$

$$\dot{y} \ddot{y} \ \dot{p} \ddot{A} \ \ddot{y} \ddot{a}$$

12. If $\ddot{y} = \dot{y} \ddot{y} + \dot{p} \ddot{A} + \ddot{y} \ddot{a}$. Evaluate $' 2 \log \dot{p} \ \ddot{y}$ (a) $\dot{y} \ddot{y} + \dot{p} \ddot{A} + \ddot{y} \ddot{a}$ (b) $\dot{y} \ddot{y} \ 2 \ \dot{p} \ddot{A} + \ddot{y} \ddot{a}$ (c) $\ddot{y}^2 + \ddot{y}^2 + \ddot{y}^2$ (d)

$$\dot{y} \ddot{y} \ \dot{p} \ddot{A} \ \ddot{y} \ddot{a}$$

$$\ddot{y}$$

$$+ \ddot{y} + \ddot{y}$$

13. Find a unit normal to the surface $\dot{y} \ 2 + \dot{p} \ 2 + \ddot{y} \ 2 \ 2 \ 4\dot{y} \dot{p} \ 2 \ 4\dot{p} \ddot{y} + 4\dot{y} \ddot{y} = 27$ $\ddot{y} \ \ddot{a} / \ddot{a} \ \dot{C} \dot{y} \dot{A} \ddot{a}$ (2, 1, 22)

$$\text{(a) } 28\ddot{y} + \ddot{A} \text{ (b) } 8\ddot{y} + \ddot{A} \text{ (c) } (2\dot{y} \ 2 \ 4\dot{p} + 4\ddot{y}) \ddot{y} + (2\dot{p} \ 2 \ 4\dot{y} \ 2 \ 4\ddot{y}) \ddot{A} + (2\ddot{y} \ 2 \ 4\dot{p} + 4\dot{y}) \ddot{a} \text{ (d) } \dot{y} \ 2 +$$

$$\dot{p} \ 2 + \ddot{y} \ 2 \ 2 \ 4\dot{y} \dot{p} \ 2 \ 4\dot{p} \ddot{y} + 4\dot{y} \ddot{y}$$

$$\dot{y} \ddot{y} + \dot{p} \ddot{A} + \ddot{y} \ddot{a}$$

14. If $' \frac{d}{dt} \dot{y} = 0$. $\frac{d}{dt} \dot{y} / \ddot{a} \ \dot{y} = \ddot{y}^3$

$\ddot{y} \ddot{A}$ & (a) Parallel (b) perpendicular (c) solenoidal (d) irrational

$$4 \ 2 \ 3 \ 2 \ 2 \ 2$$

15. If $\dot{y} = 3\dot{p} \ \ddot{y} \ \ddot{y} + 4\dot{y} \ \ddot{y} \ \ddot{A} + 3\dot{y} \ \dot{p} \ \ddot{a}$. $\frac{d}{dt} \dot{y} / \ddot{a} \ \dot{y} * \dot{y} = 0$ $\ddot{y} \ddot{A} \ \ddot{a} \ddot{y} \ddot{a}$ & ($\ddot{y}$) Parallel (b) perpendicular (c)

solenoidal (d) irrational

$$\dot{y}^2 \ \dot{y}$$

16. If $\dot{y} = (3\ddot{A}^2 + 4)\ddot{y} + (2\ddot{A} \ 2 \ 5)\ddot{A} + 4\ddot{A}^2 \ \ddot{a}$, $A\ddot{y} \ddot{A} \ \dot{y} \ddot{A}^2 \ |$. ($\ddot{y}$) 38.12 ($\ddot{A}$) 48.37 ($\ddot{a}$) 50.13 ($\ddot{A}$) 27.73

17. If $\frac{d}{dt} \dot{y} = \ddot{y} \ddot{A} \ddot{y} \ \ddot{A} \ 2 \ \ddot{a} + \ddot{A} \ \ddot{a} \ 3 \ \ddot{a} + \ddot{a} (\ddot{a} \ 3 \ 2$

$$\dot{y} \ 2 \ \frac{d}{dt} \dot{y}$$

$$4\ddot{a}), A\ddot{y} \ddot{A} \ \dot{y} \ \ddot{a} \ 2 \ |$$
. ($\ddot{y}$) 100.18 ($\ddot{A}$) 180.90 ($\ddot{a}$) 150.45 ($\ddot{A}$) 160.22

$$\ddot{a} = 1$$

18. A particle moves in speed S at time t, its position is stated as $\dot{y} = 2\ddot{a} + 3$, $\dot{p} = \ddot{a} \ 2 + 3\ddot{a}$, $\ddot{y} = \ddot{a} \ 3$ +

$$2\ddot{a} \ 2$$
. Find the velocity in the direction of the vector $2\ddot{y} + 3\ddot{A} + 4\ddot{a}$ when $\ddot{a} = 1$. (a) $4\ddot{y} + 3\ddot{A} +$

$$5\ddot{a} \ (\ddot{A}) \ 2\ddot{y} + 5\ddot{A} + 7\ddot{a} \ (\ddot{a}) \ 3\ddot{y} + 5\ddot{A} + 0\ddot{a} \ (\ddot{A}) \ 2\ddot{y} \ 2 \ 3\ddot{A} + 7 \frac{d}{dt} \dot{y}$$

19. Determine the unit tangent vector for the curve $\dot{y} = 3\ddot{a}$, $\dot{p} = 2\ddot{a} \ 2$, $\ddot{y} = \ddot{a} \ 2 + \ddot{a}$ at the point (6, 8, 6).

$$1 \ 1 \ 1 \ 1$$

$$(\ddot{y}) \ (3\ddot{y} \ 2 \ 8\ddot{A} + 5\ddot{a}) \ (\ddot{A}) \ (3\ddot{y} + 8\ddot{A} + 5\ddot{a}) \ (\ddot{a}) \ (3\ddot{y} \ 2 \ 8\ddot{A} \ 2 \ 5\ddot{a}) \ (\ddot{A}) \ (23\ddot{y} \ 2 \ 8\ddot{A} + 5\ddot{a})$$

$$: 98 : 98 : 98 : 98$$

$$2 \ 2$$

20. If $\frac{d}{dt} \dot{y} = 3\ddot{A} \ddot{y} + \ddot{A} \ \ddot{A} + (\ddot{A} + 2)\ddot{a}$, $\frac{d}{dt} \dot{y} = 2\ddot{A} \ddot{y} \ 2 \ 3\ddot{A} \ddot{A} + (\ddot{A} \ 2 \ 2)\ddot{a}$, $\ddot{a} \ddot{y} \ddot{A} \ddot{y} \ddot{a} \ \int 0 \ (\frac{d}{dt} \dot{y} \times \frac{d}{dt} \dot{y}) \ddot{A} \ddot{A}$.

$$4 \ 4 \ 4 \ 4$$

$$(\ddot{y}) \ (214\ddot{y} + 13\ddot{A} \ 2 \ 24\ddot{a}) \ (\ddot{A}) \ (14\ddot{y} + 13\ddot{A} \ 2 \ 24\ddot{a}) \ (\ddot{a}) \ (14\ddot{y} \ 2 \ 13\ddot{A} \ 2 \ 24\ddot{a}) \ (\ddot{A}) \ (14\ddot{y} +$$

3 3 3 3

$13\bar{A} + 24\bar{a}$)

21. If the point (2, k) lies on the line with slope $m = 3$ passing through the point (1, 6), find k. (a) 12 (b).

4 (c) 9 (d) -9

22. Find the slope m of the line having the equation $y - 7 = 2(x - 3)$ and find a point on the line. (a) m

= -2 & (7, -3) (b) $m = 2$ & (3, 7) (c) $m = 3$ & (-2, 7) (d) $m = 7$ & (2, 3)

$\bar{y} \bar{p} \bar{y} \bar{p} \bar{y}$

23. Find an equation of the line through the points (0,2) and (3,0). (a) $\bar{y} + 2 = 1$ (b) $2 + 3 = 1$ (c) $3 + 3$

$1 \bar{y} \bar{p}$

$\bar{p} = (d) + = 0$

2 3 2

24. Find an equation of the line through the point (2, -3) and perpendicular to the line $4x - 5y = 7$. (a)

$2\bar{y} + 3\bar{p} = 5$ (b) $4\bar{y} + 2\bar{p} = 3$ (c) $3\bar{y} + 2\bar{p} = 24$ (d) $5\bar{y} + 4\bar{p} = 22$

25. For what value of k will the line $kx + 5y - 2k$ have slope 3? (a) 12 (b) -15 (c) -9 (d) -10

26. Write the standard equation for the circle with center (3,5) and radius 4. (a) $(\bar{y} + 5)^2 + (\bar{y} + 3)^2 =$

4 (b) $(\bar{y} - 5)^2 + (\bar{y} - 3)^2 = 16$ (c) $(\bar{y} - 3)^2 + (\bar{y} - 5)^2 = 16$ (d) $(\bar{y} - 3)^2 + (\bar{y} - 5)^2 = 4$

3 9

27. Given $\bar{y} = :11$ and centre (0, 2), find the equation of the circle. (a) $22\bar{y}^2 + 22\bar{p}^2 + 18\bar{p} + 9 = 22$

0 (b) $2\bar{y}^2 + 2\bar{p}^2 + 11\bar{p} + 9 = 0$ (c) $22\bar{y}^2 + 22\bar{p}^2 + 11\bar{p} + 9 = 0$ (d) $2\bar{y}^2 + 2\bar{p}^2 + 18\bar{p} + 9 = 0$

28. The diameter of a circle has its coordinates as A(-6, -1) and B(16, 13). Find the centre and radius

(a) (5, 6), $r = 10$ (b) (5, 6), $r = :55$ (c) (6, 5), $r = 55$ (d) (-5, -6), $r = :55$

2 5 2 5

29. Let the parabola be $4\bar{y} + 10\bar{p}^2 = 5\bar{p} + 5$. Find latus rectum (a) 2 3 (b) 2 (c) 2 (d) 2

8 5 2

2 5 2 5

30. Let the parabola be $4\bar{y} + 10\bar{p}^2 = 5\bar{p} + 5$. Find length of the latus rectum. (a) 3 (b) 8 (c) 5 (d) 2 1 45 45 1 1 9 1 9

31. Let the parabola be $4\bar{y} + 10\bar{p}^2 = 5\bar{p} + 5$. Find vertex. (a) (,

) (b) (,) (c) (,) (d) (,)

4 32 32 4 2 16 4 16

1 1 45 45 1 1 1

32. Let the parabola be $4\bar{y} + 10\bar{p}^2 = 5\bar{p} + 5$. Find focus (a) (4 2 10 , 32) (b) (32 2 10 , 4) (c) (2 2 1 9 1 1 9

,) (d) (2 ,)

10 16 4 10 16

2

33. Let the parabola be $4y^2 + 10p = 5p + 5$. Find equation of the directrix. (a) $160y^2 - 241 = 0$ (b) $241p^2 - 160 = 0$ (c) $241y^2 - 160 = 0$ (d) $160p^2 - 241 = 0$

34. Which of the following is not an example of projectile motion? (a) a hot air balloon drifting towards

earth. (b) a volleyball served over a net (c) a long jumper in action (d) a baseball hit by a bat.

35. Lisa goes into a jump at 12m/s and launches herself at 20% above the horizontal. How long is she

in the air before returning to earth? ($g = 9.81 \text{ m/s}^2$). (a) 1.5 s (b) 1.2 s (c) 0.42 s (d) 0.83 s

36. What is the path of a projectile? (a) a hyperbola (b) projectiles do not follow a predictable path (c)

a wavy line (d) a parabola

37. A body of mass 10 kg, projected at an angle of 30° from the ground with an initial velocity of 5m/s,

acceleration due to gravity is $a = 10 \text{ m/s}^2$, what is the time of flight? (a) 0.866 s (b) 1.86 s (c) 1.96 s (d) 1.862 s

38. When do we get maximum height in a simple projectile motion? (a) When $\theta = 45^\circ$ (b) When $\theta =$

60° (c) When $\theta = 90^\circ$ (d) When $\theta = 0^\circ$

39. A soccer ball is projected at an angle of 60° from the ground. It attains its maximum height in 10s.

Considering acceleration due to gravity as $a = 10 \text{ m/s}^2$. What is the velocity with which it was projected? (a) 11.55 m/s (b) 11.5 m/s (c) 1.155 m/s (d) 12.0 m/s

40. An elastic string of natural length 5Å and modulus μ is cut into parts of length 2Å and 3Å. Each

part is tied to the same particle of mass m. Their free ends are tied to two fixed points a distance 7Å

apart on a smooth horizontal table. Find the period of oscillation when the particle is displaced

$5\mu/7\mu$ 3Å 6Å

along the string. (a) (b) (c) $2\mu/\sqrt{}$ (d) $2\mu/\sqrt{}$

$6\mu/3\mu$ 7Å 5Å

41. Two spheres of masses 3 kg and 4 kg on a smooth horizontal table are connected with a light inextensible string of length a. Initially the string is slack and the 4 kg mass is projected along the table with velocity 21 m/s. Find the velocity with which 3 kg mass begins to move. (a) 7 m/s (b) 12 m/s (c) 17 m/s (d) 5 m/s

42. Two spheres of masses 3 kg and 4 kg on a smooth horizontal table are connected with a light inextensible string of length a. Initially the string is slack and the 4 kg mass is projected along the table with velocity 21 m/s. Find the impulse generated in the string when it suddenly becomes taut. (a) 36 Newton-sec (b) 12 Newton-sec (c) 6 Newton-sec (d) 10 Newton-sec

43. Find the length of a simple pendulum if its period for small oscillations is 1 sec. Take acceleration

due to gravity as 10 m/s^2 . (a) 0.253 m (b) 0.005 m (c) 25 m (d) none of the above

44. If the length of a simple pendulum is increased by 1% by what percentage is its period for small

oscillation increased. (a) 1 % (b) 0.4988 % (c) 50% (d) none of the above

45. The length of a simple pendulum is 0.25 m and its maximum angle of swing is 8° . What is the frequency of oscillation? (a) 7 Hz (b) 2.008 Hz (c) 3.667 Hz (d) 1.007 Hz

46. A body of mass 4 kg is hung on a light spring and is found to stretch it 3 cm. The mass is the pulled

down a further 2 cm and released. Find the period of oscillation of the mass. (a) 0.3444 sec (b) 0.333 sec (c) 0.4555 sec (d) 7 sec

47. A body of mass 4 kg is hung on a light spring and is found to stretch it 3 cm. The mass is the pulled

down a further 2 cm and released. Find its maximum speed. (a) 0.5 m/s (b) 0.72 m/s (c) 0.365 m/s (d) 1.5 m/s

48. An elastic spring extends a distance $\check{A}$ when a given mass is attached to its lowest point. The mass

is pulled down a further distance $\check{y} (< \check{A})$ and let go. Find the period of the simple harmonic motion $\check{a}$

that ensues. (a) $\check{y}\sqrt{\check{A}}$ (b) $2\sqrt{\check{A}\check{a}}$ (c) $2\sqrt{\check{y}\check{A}\check{a}}$ (d) none of the above

49. A truck with a momentum of 85000 kgm/s is moving with a velocity of 50 m/s. Determine its mass

(a) 1700 $\check{a}\check{q}$ (b) 850 $\check{a}\check{q}$ (c) 70 $\check{a}\check{q}$ (d) 180 $\check{a}\check{q}$

50. What force will give a mass of 8kg a velocity of $15\check{a}/\check{A}$ in 1 minute starting from rest. (a) $2\check{a}\check{q}$ (b) $5\check{a}\check{q}$

(c) $7.5\check{a}\check{q}$ (d) $12\check{a}\check{q}$

51. Find the acceleration produced when a force of 5N acts on a mass of 10kg (a) 12m/s (b) 7m/s (c) 0.5m/s

(d) 3m/s

52. A particle moves along a curve C such that its position vector is given by $\check{y} = 5\check{a}\check{y} + 3\check{a}^2\check{A} + 5\check{a}^3\check{a}$.

Find its acceleration at the anytime t. (a) $6\check{A} + 30\check{a}\check{a}$ (b) $5\check{y} + 6\check{a}\check{A} + 15\check{a}^2\check{a}$ (c) $5\check{a}\check{y} + 3\check{a}^2\check{A} + 5\check{a}^3\check{a}$

(d) 6j

53. Given a space curve C with position vector $\check{y} = 2\check{a}\check{y} + 6\check{a}\check{A} + 9\check{a}\check{a}$. Find the radius of curvature (a)

1 25

11 (b) 11 (c) 3 (d) 10

54. Find the unit principal normal of $\check{y} = 2\check{a}\check{y} + 6\check{a}\check{A} + 9\check{a}\check{a}$.

1

(a) $11(2\check{y} + 6\check{A} + 9\check{a})$ (b) $11(2\check{y} + 6\check{A} + 9\check{a})$ (c) $2\check{y} + 6\check{A} + 9\check{a}$ (d) $2\cos 2\check{a}\check{y}^2\sin 2\check{a}\check{A}$

55. A particle moves with a position vector $\check{y}\check{a}\check{q}\check{A} + \sin\check{a}\check{y} + \sin\check{a}\check{A}$. What is the value for which the velocity

of the particle is perpendicular to ? (a) 1 (b) 0 (c) $2r$ (d) k

56. Suppose particle P moves on a circle C of radius R. If s is the length of arc measures along C from

A to P and $\check{a}$ is the corresponding angle subtended at the centre O. Find the normal acceleration of

$$\vec{r} = \vec{r}_1 + \vec{r}_2 + \vec{r}_3$$

the body if v is the tangential velocity. (a) $\vec{r}_1$ (b) $\vec{r}_2$ (c) $\vec{r}_3$ (d) $\vec{r}_4$

57. Find the sum of the vectors: $4\vec{OA}$, $3\vec{AO}$, $\vec{AB}$, $\vec{BZ}$, $7\vec{OB}$, $7\vec{BZ}$. (a) $5\vec{OZ}$ (b) $6\vec{OZ}$ (c) $7\vec{OZ}$ (d) $8\vec{OZ}$

58. Given $\vec{y}_1 = 3\vec{y}_2 + 2\vec{A} + \vec{a}$, $\vec{y}_2 = 2\vec{y}_3 + 4\vec{A} + 3\vec{a}$, $\vec{y}_3 = 2\vec{y}_1 + 2\vec{A} + 2\vec{a}$. Find $|2\vec{y}_1 + 3\vec{y}_2 + 5\vec{y}_3|$ (a) 30 (b) 40 (c) 3 (d) 5

59. Find the work done in moving an object along a vector $\vec{y} = 3\vec{y}_2 + 2\vec{A} + 5\vec{a}$ if the applied force is $\vec{F} =$

$$2\vec{y}_2 + \vec{A} + \vec{a}$$

$$2\vec{y}_2 + \vec{A} + \vec{a}$$

(a) 8 (b) 9 (c) 7 (d) 6

60. Find the angle between the vectors $\vec{y}_1 = 3\vec{y}_2 + 2\vec{A} + 6\vec{a}$ & $\vec{y}_2 = 4\vec{y}_3 + 3\vec{A} + \vec{a}$ (a) 57° (b) 79° (c) 90° (d) 69°

(d) 69°

61. Find the $\vec{y} \cdot \vec{a}$ if $\vec{y} = 4\vec{y}_2 + 3\vec{A} + \vec{a}$ and $\vec{a} = 22\vec{y}_3 + 4\vec{A} + 3\vec{a}$ (a) -23 (b) -21 (c) 220 (d) -12

62. Speed of a truck is 40 ms^{-1} , after 10 seconds its speed decreases to 20 ms^{-1} , its acceleration is (a) -1 ms^{-2} (b) -2 ms^{-2} (c) -4 ms^{-2} (d) -5 ms^{-2}

1

(b) -2 ms^{-2} (c) -4 ms^{-2} (d) -5 ms^{-2}

63. A car stops and then starts accelerating uniformly at rate of 3 ms^{-2} . Speed of car after 20 seconds is (a) 40 ms^{-2} (b) 30 ms^{-2} (c) 100 ms^{-2} (d) 60 ms^{-2}

40 ms^{-2} (b) 30 ms^{-2} (c) 100 ms^{-2} (d) 60 ms^{-2}

64. A stone is thrown upwards with initial velocity of 20 ms^{-1} , height that stone will reach would be (a) 20 m (b) 30 m (c) 40 m (d) 50 m

m (b) 30 m (c) 40 m (d) 50 m

65. If initial velocity of an object is zero, then distance covered by it in time t and acceleration of 9.8 ms^{-2} is (a) $2.9 t^2$ (b) $3 t^2$ (c) $4 t^2$ (d) $4.9 t^2$

is (a) $2.9 t^2$ (b) $3 t^2$ (c) $4 t^2$ (d) $4.9 t^2$

66. Horizontal distance travelled by a ball if it's thrown with initial velocity of 20 ms^{-1} at an angle of 30° is (a) 24 m (b) 56 m (c) 35.3 m (d) 36.3 m

(a) 24 m (b) 56 m (c) 35.3 m (d) 36.3 m

67. Find the midpoint of the line segment joining the points $\vec{y}(4, 22)$ & $\vec{y}(28, 6)$ (a) (6,4) (b) (-6,4) (c) (2,2) (d) (-2,2)

5 3

68. Find the distance between the points $(-3, 2)$ and $(6, 4)$ (a) :85 (b) :79 (c) :3 (d) :5

4 3 8 5

69. What is the gradient of the line passing through the points $(4, 6)$ and $(-1, -2)$? (a) 3 (b) 4 (c) 5 (d) 8

70. Which points satisfies the linear quadratic system $p = y + 3$ and $p = 5 + 2y$? (a) $(-2, 1)$ (b) $(2, 1)$ (c) $(-1, 2)$ (d) $(4, -1)$

1,2) (d) (4,-1)

Paper 2 of 4 — MTH 103 or MTH 123 Vector And Dynamics Second Semester 2020 2021

Level 100 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN
COLLEGE OF SCIENCE
DEPARTMENT OF MATHEMATICS

SECOND SEMESTER EXAMINATION 2020/2021

Course code & title: MTH 123 – Vectors & Dynamics. Duration: 2HRS MINS
 TYPE D: Instruction: Answer all questions.

- When do we get maximum height in a simple projectile motion? (a) When $\theta = 45^\circ$ (b) When $\theta = 60^\circ$ (c) When $\theta = 90^\circ$ (d) When $\theta = 0^\circ$
- A soccer ball is projected at an angle of 60° from the ground. It attains its maximum height in 10s. Considering acceleration due to gravity as $g = 10 \text{ m/s}^2$. What is the velocity with which it was projected? (a) 11.55 m/s (b) 11.5 m/s (c) 1.155 m/s (d) 12.0 m/s
- Two spheres of masses 3 kg and 4 kg on a smooth horizontal table are connected with a light inextensible string of length a . Initially the string is slack and the 4 kg mass is projected along the table with velocity 21 m/s. Find the velocity with which 3 kg mass begins to move. (a) 7 m/s (b) 12 m/s (c) 17 m/s (d) 5 m/s
- Two spheres of masses 3 kg and 4 kg on a smooth horizontal table are connected with a light inextensible string of length a . Initially the string is slack and the 4 kg mass is projected along the table with velocity 21 m/s. Find the impulse generated in the string when it suddenly becomes taut. (a) 36 Newton-sec (b) 12 Newton-sec (c) 6 Newton-sec (d) 10 Newton-sec
- Find the length of a simple pendulum if its period for small oscillations is 1 sec. Take acceleration due to gravity as 10 m/s^2 . (a) 0.253 m (b) 0.005 m (c) 25 m (d) none of the above
- If the length of a simple pendulum is increased by 1% by what percentage is its period for small oscillation increased. (a) 1% (b) 0.4988% (c) 50% (d) none of the above
- The length of a simple pendulum is 0.25 m and its maximum angle of swing is 8° . What is the frequency of oscillation? (a) 7 Hz (b) 2.008 Hz (c) 3.667 Hz (d) 1.007 Hz
- A body of mass 4 kg is hung on a light spring and is found to stretch it 3 cm. The mass is pulled down a further 2 cm and released. Find the period of oscillation of the mass. (a) 0.3444 sec (b) 0.333 sec (c) 0.4555 sec (d) 7 sec
- A body of mass 4 kg is hung on a light spring and is found to stretch it 3 cm. The mass is pulled down a further 2 cm and released. Find its maximum speed. (a) 0.5 m/s (b) 0.72 m/s (c) 0.365 m/s (d) 1.5 m/s
- A particle moves along a curve C such that its position vector is given by $r = 5ti + 3t^2j + 5t^3k$. Find its acceleration at the anytime t . (a) $6j + 30tk$ (b) $5i + 6tj + 15t^2k$ (c) $5ti + 3t^2j + 5t^3k$ (d) $6j$
- Given a space curve C with position vector $r = 2ti + 6tj + 9tk$. Find the radius of curvature (a) 11 (b) $\frac{1}{11}$ (c) $\frac{25}{3}$ (d) 10
- Find the unit principal normal of $r = 2ti + 6tj + 9tk$. (a) $\frac{1}{11}(2i + 6j + 9k)$ (b) $11(2i + 6j + 9k)$ (c) $2i + 6j + 9k$ (d) $-\cos 2ti - \sin 2tj$
- A particle moves with a position vector $r = \cos \theta i + \sin \theta j$. What is the value for which the velocity of the particle is perpendicular to r ? (a) 1 (b) 0 (c) $-\theta^2 r$ (d) θk
- Suppose particle P moves on a circle C of radius R . If s is the length of arc measures along C from A to P and θ is the corresponding angle subtended at the centre O . Find the normal acceleration of the body if v is the tangential velocity. (a) $\frac{d^2\theta}{dt^2}$ (b) $\frac{d^2s}{dt^2}$ (c) $\frac{v^2}{R}$ (d) $\frac{v^2}{R}$
- The numerical value of a vector is known as a --? (a) magnitude (b) position vector (c) unit vector (d) null vector
- The position vector at origin is equivalent to (a) unit vector (b) equal vector (c) null vector (d) all of the above
- What type of vector is only of mathematical importance? (a) unit vector (b) equal vector (c) null vector (d) none
- What is the absolute value of $\frac{\vec{A}}{|\vec{A}|}$ (a) 0 (b) 2 (c) $\vec{A}$ (d) 1
- Vectors having the same initial point are called (a) concurrent (b) coplanar (c) parallel (d) all of the above
- Orthogonal vectors have angle equal to (a) 60 degree (b) 90 degree (c) 120 degree (d) 180 degree
- In vector cross product, $b \times a$ is equivalent to (a) $-a \times b$ (b) $a \times b$ (c) $-b \times a$ (d) $a \times a$
- The result of dot product of two vectors is -- (a) vector (b) scalar (c) direction (d) all of the above
- Given that $\vec{A} = 2i + j + 2k$. Find the square of $|\vec{A}|$. (a) 3 (b) 4 (c) 9 (d) 10
- Given $\vec{A} = 4i + j + 3k$ and $\vec{B} = -4i + 2j + 8k$. Obtain $\vec{A} \cdot \vec{B}$. (a) 10 (b) 26 (c) 22.8 (d) 42
- Given that $\vec{A} = 2i - 3j + 5k$ and $\vec{B} = -6i + 9j - 15k$. calculate the angle between them. (a) 2π (b) 60 (c) π (d) 3π
- A function of two vectors such that $u(A) \cdot u(B) = 0$ is (a) Parallel (b) unit vector (c) perpendicular (d) scalar triple product
- A continuous vector is --- (a) $\lim_{u \rightarrow \Delta u} \frac{A(u+\Delta u) - A(u)}{\Delta u}$ (b) $\lim_{u \rightarrow \Delta u} \frac{A(u+\Delta u) - A(u)}{\Delta u + u}$ (c) $\lim_{u \rightarrow u_0} A(u) = A(u_0)$ (d) $\lim_{u \rightarrow \Delta u} A(u) = A(u_0)$
- Express $\frac{d}{dx}(AXB)$ as a product. (a) $A \frac{dB}{dx} + \frac{dA}{dx} XB$ (b) $A * \frac{dB}{dx} + \frac{dA}{dx} * B$ (c) $A \frac{\partial B}{\partial x} + \frac{\partial A}{\partial x} XB$ (d) $A * \frac{\partial B}{\partial x} + \frac{\partial A}{\partial x} * B$
- Find an equation of the line through the point (2, -3) and perpendicular to the line $4x - 5y = 7$. (a) $2x + 3y = 5$ (b) $4x + 2y = 3$ (c) $3x + 2y = -4$ (d) $5x + 4y = -2$
- For what value of k will the line $kx + 5y - 2k$ have slope 3? (a) 12 (b) -15 (c) -9 (d) -10

38. Given $\hat{A} = 4i + j + 3k$ and $\hat{B} = -4i + 2j + 8k$. Obtain $\hat{A} \cdot \hat{B}$. (a) 10 (b) 26 (c) 22.8 (d) 42
39. Given that $\hat{A} = 2i - 3j + 5k$ and $\hat{B} = -6i + 9j - 15k$. calculate the angle between them. (a) 2π (b) 60 (c) π (d) 3π
40. A function of two vectors such that $u(A) \cdot u(B) = 0$ is (a) Parallel (b) unit vector (c) perpendicular (d) scalar triple product
41. A continuous vector is--- (a) $\lim_{u \rightarrow \Delta u} \frac{A(u+\Delta u) - A(u)}{\Delta u}$ (b) $\lim_{u \rightarrow \Delta u} \frac{A(u+\Delta u) - A(u)}{\Delta u + u}$ (c) $\lim_{u \rightarrow u_0} A(u) = A(u_0)$ (d) $\lim_{u \rightarrow \Delta u} A(u) = A(u_0)$
42. Express $\frac{d}{dx}(AXB)$ as a product. (a) $AX \frac{dB}{dx} + \frac{dA}{dx}XB$ (b) $A * \frac{dB}{dx} + \frac{dA}{dx} * B$ (c) $AX \frac{\partial B}{\partial x} + \frac{\partial A}{\partial x}XB$ (d) $A * \frac{\partial B}{\partial x} + \frac{\partial A}{\partial x} * B$
43. Given that $\phi(x, y, z) = x^2yz$. Find $\nabla\phi$. (a) $yz^2i + 2xyzk$ (b) $2xyi + x^2zj + x^2yk$ (c) $2yzi + x^2yk$ (d) $2xyz + x^2zj + x^2yk$
44. If $r = xi + yj + zk$. Evaluate $\nabla^2 \log_e r$ (a) $xi + yj + zk$ (b) $xi - yj + zk$ (c) $\frac{xi}{r^2} + \frac{yj}{r^2} + \frac{zk}{r^2}$ (d) $\frac{xi}{r} + \frac{yj}{r} + \frac{zk}{r}$
45. If $\nabla XA = 0$. Then $A = \frac{xi+yj+zk}{r^3}$ is... (a) Parallel (b) perpendicular (c) solenoidal (d) irrational
46. If $A = 3y^4z^2i + 4x^3z^2j + 3x^2y^2k$. Then $\nabla * A = 0$ is term... (a) Parallel (b) perpendicular (c) solenoidal (d) irrational
47. If $A = (3u^2 + 4)i + (2u - 5)j + 4u^2k$, find $\left| \frac{d^2A}{du^2} \right|$. (a) 38.12 (b) 48.37 (c) 50.13 (d) 27.73
48. If $F = i \sin 2t + je^{3t} + k(t^3 - 4t)$, find $\left| \frac{d^2F}{dt^2} \right|_{t=1}$. (a) 100.18 (b) 180.90 (c) 150.45 (d) 160.22
49. Given $\hat{A} = 3i + 4j - 7k$, $\hat{B} = -i + 2j + k$ and $C = i + 2k$. Obtain $\hat{A} \times \hat{B}$. (a) (10, 5, 7) (b) (6, 2, 4) (c) (18, 4, 10) (d) (1, 5, 8)
50. Given $\hat{A} = 3i + 4j - 7k$, $\hat{B} = -i + 2j + k$ and $C = i + 2k$. Obtain $(\hat{A} \times \hat{B}) \cdot C$ (a) 38 (b) 20 (c) 35 (d) 40
51. Given $\hat{A} = 3i + 4j - 7k$, $\hat{B} = -i + 2j + k$ and $C = i + 2k$. Obtain $(\hat{A} \times \hat{B}) \times C$ (a) (1, 5, 7) (b) (8, -26, -4) (c) (8, 4, 10) (d) (8, 5, 8)
52. Given $\hat{A} = 3i + 4j - 7k$, $\hat{B} = -i + 2j + k$ and $C = i + 2k$. Obtain $\hat{A} \times (\hat{B} \times C)$ (a) (0, 5, 8) (b) (12, -2, 4) (c) (3, 3, 0) (d) (13, -22, -7)
53. If $u(t) = (t^2 + 1, \sin 3\pi t, t)$ find $\lim_{t \rightarrow 2} u(t)$ (a) (5, 0, 2) (b) (5, -1, 0) (c) (-2, 0, 2) (d) (1, 0, -7)
54. If the point (2, k) lies on the line with slope $m = 3$ passing through the point (1, 6), find k. (a) 12 (b) 4 (c) 9 (d) -9
55. Find the slope m of the line having the equation $y - 7 = 2(x - 3)$ and find a point on the line. (a) $m = -2$ & (7, -3) (b) $m = 2$ & (3, 7) (c) $m = 3$ & (-2, 7) (d) $m = 7$ & (2, 3)
56. Find an equation of the line through the points (0, 2) and (3, 0). (a) $\frac{x}{3} + \frac{y}{2} = 1$ (b) $\frac{x}{2} + \frac{y}{3} = 1$ (c) $\frac{x}{3} + y = \frac{1}{2}$ (d) $\frac{x}{3} + \frac{y}{2} = 0$
57. Find an equation of the line through the point (2, -3) and perpendicular to the line $4x - 5y = 7$. (a) $2x + 3y = 5$ (b) $4x + 2y = 3$ (c) $3x + 2y = -4$ (d) $5x + 4y = -2$
58. For what value of k will the line $kx + 5y - 2k$ have slope 3? (a) 12 (b) -15 (c) -9 (d) -10
59. Write the standard equation for the circle with center (3, 5) and radius 4. (a) $(x + 5)^2 + (x + 3)^2 = 4$ (b) $(x - 5)^2 + (x - 3)^2 = 16$ (c) $(x - 3)^2 + (x - 5)^2 = 16$ (d) $(x - 3)^2 + (x - 5)^2 = 4$
60. Given $r = \frac{3}{2}\sqrt{11}$ and centre (0, $-\frac{9}{2}$), find the equation of the circle. (a) $-2x^2 - 2y^2 - 18y + 9 = 0$ (b) $2x^2 + 2y^2 - 11y + 9 = 0$ (c) $-2x^2 - 2y^2 - 11y + 9 = 0$ (d) $2x^2 - 2y^2 - 18y + 9 = 0$
61. The diameter of a circle has its coordinates as A(-6, -1) and B(16, 13). Find the centre and radius (a) (5, 6), $r = 10$ (b) (5, 6), $r = \sqrt{55}$ (c) (6, 5), $r = 55$ (d) (-5, -6), $r = \sqrt{55}$
52. Let the parabola be $4x + 10y^2 = 5y + 5$. Find latus rectum (a) $-\frac{2}{3}$ (b) $-\frac{5}{8}$ (c) $-\frac{2}{5}$ (d) $-\frac{5}{2}$
53. Let the parabola be $4x + 10y^2 = 5y + 5$. Find length of the latus rectum. (a) $\frac{2}{3}$ (b) $\frac{5}{8}$ (c) $\frac{2}{5}$ (d) $\frac{5}{2}$
54. Let the parabola be $4x + 10y^2 = 5y + 5$. Find vertex. (a) $(\frac{1}{4}, \frac{45}{32})$ (b) $(\frac{45}{32}, \frac{1}{4})$ (c) $(\frac{1}{2}, \frac{9}{16})$ (d) $(\frac{1}{4}, \frac{9}{16})$
55. Let the parabola be $4x + 10y^2 = 5y + 5$. Find focus (a) $(\frac{1}{4} - \frac{1}{10}, \frac{45}{32})$ (b) $(\frac{45}{32} - \frac{1}{10}, \frac{1}{4})$ (c) $(\frac{1}{2} - \frac{1}{10}, \frac{9}{16})$ (d) $(\frac{1}{4} - \frac{1}{10}, \frac{9}{16})$
56. Let the parabola be $4x + 10y^2 = 5y + 5$. Find equation of the directrix. (a) $160x - 241 = 0$ (b) $241y - 160 = 0$ (c) $241x - 160 = 0$ (d) $160y - 241 = 0$
57. Which of the following is not an example of projectile motion? (a) a hot air balloon drifting towards earth. (b) a volleyball served over a net (c) a long jumper in action (d) a baseball hit by a bat.
58. Lisa goes into a jump at 12m/s and launches herself at 20% above the horizontal. How long is she in the air before returning to earth? ($g = 9.81 \text{ m/s}^2$). (a) 1.5 s (b) 1.2 s (c) 0.42 s (d) 0.83 s
59. What is the path of a projectile? (a) a hyperbola (b) projectiles do not follow a predictable path (c) a wavy line (d) a parabola
60. A body of mass 10 kg, projected at an angle of 30° from the ground with an initial velocity of 5m/s, acceleration due to gravity is $g = 10 \text{ m/s}^2$, what is the time of flight? (a) 0.866 s (b) 1.86 s (c) 1.96 s (d) 1.862 s

Paper 3 of 4 — MTH 103 VECTORS & DYNAMICS 2018,2019

Level 100 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURU
COLLEGE OF SCIENCE
DEPARTMENT OF MATHEMATICS /COMPUTER SCIENCE
SECOND SEMESTER EXAMINATION 2018/2019

Course code & title: MTH 123 – Vectors & Dynamics Duration: 1HR 45MINS

Instruction: Answer all questions. Submit your question paper and answer booklet at the end of the exam

TYPE B

1. Find the equation of the tangent to the curve $y = 2x^3 - x^2 + 3x + 1$ at the point $x = 1$
 (a) $7x - 2$ (b) $7x + 2$ (c) $7x - 7$ (d) $6x^2 - 2x + 3$
2. Find the equation of the normal to the curve $y = x + \frac{1}{x}$ at the point $x = 2$
 (a) $(2, 2)$, $(-1, -2)$ (b) $(-2, -2)$, $(2, 2)$ (c) $x - 2$ (d) ± 2
3. What is the equation of the normal to the curve $f(x) = \frac{x^3}{3} + x^2 + x - \frac{1}{3}$ which is parallel to the line $y = -\frac{1}{4}x + \frac{1}{3}$
 (a) $\frac{9-x}{4}$, $\frac{3x-49}{12}$ (b) $\frac{9+x}{4}$, $\frac{3x-49}{12}$ (c) $\frac{9+x}{4}$, $\frac{3x+49}{12}$ (d) $-\frac{1}{4}x + \frac{1}{3}$, $\frac{9-x}{4}$
4. An 80-kg person weighs 765-N on top of a mountain. Calculate the acceleration due to gravity on the mountain.
 (a) 9.56 m/s^2 (b) 11 m/s^2 (c) 19 m/s^2 (d) 12 m/s^2
5. A truck with a momentum of 85000 kgm/s is moving with a velocity of 50 m/s. Determine its mass
 (a) 1700 kg (b) 850 kg (c) 70 kg (d) 180 kg
6. What force will give a mass of 8kg a velocity of 15m/s in 1 minute starting from rest. (a) 2N (b) 5N (c) 7.5N (d) 12N
7. Find the acceleration produced when a force of 5N acts on a mass of 10kg (a) 12 m/s^2 (b) 7 m/s^2 (c) 0.5 m/s^2 (d) 3 m/s^2
8. A particle moves along a curve C such that its position vector is given by $r = 5ti + 3t^2j + 5t^3k$. Find its acceleration at the anytime t. (a) $6j + 30tk$ (b) $5i + 6tj + 15t^2k$ (c) $5ti + 3t^2j + 5t^3k$ (d) $6j$
9. Given a space curve C with position vector $r = 2ti + 6tj + 9tk$. Find the radius of curvature (a) 11 (b) $\frac{1}{11}$ (c) $\frac{25}{3}$ (d) 10
10. Find the unit principal normal of $r = 2ti + 6tj + 9tk$.
 (a) $\frac{1}{11}(2i + 6j + 9k)$ (b) $11(2i + 6j + 9k)$ (c) $2i + 6j + 9k$ (d) $-\cos 2ti - \sin 2tj$
11. If initial velocity of an object is zero, then distance covered by it in time t and acceleration of 9.8 ms^{-2} is
 (a) $2.9 t^2$ (b) $3 t^2$ (c) $4 t^2$ (d) $4.9 t^2$
12. Horizontal distance travelled by a ball if it's thrown with initial velocity of 20 ms^{-1} at an angle of 30° is.
 (a) 24 m (b) 56 m (c) 35.3 m (d) 36.3 m
13. If a projectile is launched horizontally with a speed 12m/s from the top of a 24.6m high building. Determine the horizontal displacement of the projectile. (a) 45m (b) 2.25m (c) 18m (d) 27m
14. Oblique impact of two spheres implies that the angles between the spheres are--
 (a) zeros (b) positives (c) non-negatives (d) does not exist
15. A football is kicked with an initial velocity of 25m/s at an angle 45° with the horizontal. Determine the maximum height of the football. (a) 0.54m (b) 90m (c) 48m (d) 16m
16. A ball rolls off the edge of a table. The horizontal component of the ball's velocity remains constant during its entire trajectory because... (a) the ball is not acted upon by any force (b) the net force acting on the ball is zero (c) the ball is not acted upon by a force in the horizontal direction (d) None of the above
17. A car accelerates from rest at 3 ms^{-2} for 4secs. How far has it travelled? (a) 54m (b) 24m (c) 48m (d) 16m
18. A sphere of mass 0.1kg moving with a velocity of 6m/s collides directly with a sphere of mass 0.2kg at rest. Calculate the common velocity if both spheres move off together. (a) 2m/s (b) 3m/s (c) 4m/s (d) 5m/s
19. Find the equation of the tangent to $y = \frac{1}{x-3}$, $x \neq 3$ which has slope 2. (a) $2x - y = 1$ (b) None
 (c) $y - 2x = 0$ (d) $y - 2x = 3$
20. The equation of the normal to the curve $x^2 = 4y$ which passes through the point (1,2) is
 (a) $x + 2 - 3 = 0$ (b) $4x - y = 2$ (c) $4x - 2y = 0$ (d) $4x - 3y + 2 = 0$
 Use the following to answer 21 to 27. Let the equation of the hyperbola be $4x^2 - 5y^2 - 16x + 10y + 31 = 0$. Find its
21. Coordinates of the Centre. (a) (5, 6) (b) (2, 1) (c) (-3, 4) (d) (4, 2)
22. Transverse and Conjugate axes. (a) 8 and $2\sqrt{7}$ (b) 6 and $2\sqrt{5}$ (c) 4 and $2\sqrt{7}$ (d) 4 and $2\sqrt{5}$
23. Vertices (a) $V_1(2, -1)$ & $V_2(2, 3)$ (b) $V_1(2, 1)$ & $V_2(3, 2)$ (c) $V_1(1, -2)$ & $V_2(2, 5)$ (d) $V_1(-2, -1)$ & $V_2(3, 3)$
24. Eccentricity (a) $\frac{9}{4}$ (b) $\frac{9}{2}$ (c) $\frac{3}{2}$ (d) $\frac{3}{4}$
25. Foci (a) $F_1(-4)$ & $F_2(2, 3)$ (b) $F_1(2, -2)$ & $F_2(2, 4)$ (c) $F_1(2, -1)$ & $F_2(3, 3)$ (d) $F_1(3, -1)$ & $F_2(2, 4)$
26. Latus recti (a) 1 (b) 4 (c) 3 (d) 2

27. Equation of the directrices.
 (a) $L_1(-1/3)$ & $L_2(5/3)$ (b) $L_1(2/3)$ & $L_2(7/3)$ (c) $L_1(-2/3)$ & $V_2(5/3)$ (d) $L_1(-1/3)$ & $L_2(7/3)$
28. Given the equation of the circle $x^2 + y^2 - 4x + 6y - 12 = 0$. Obtain the coordinates of the Centre and radius
 (a) (2, -3), $r = 5$ (b) (3, 5), $r = 2$ (c) (4, 5), $r = 7$ (d) (3, -4), $r = 5$
29. Find the equation of the tangent at the point $x = 0$ and $x = \pi/6$ at which $f(x) = \sin 2x$.
 (a) $y = 2x$, $y = 1 - 2x - \frac{\pi}{6}$ (b) $y = 2x$, $y = x + \frac{\sqrt{3}}{2} - \frac{\pi}{6}$ (c) $y = x + \frac{\sqrt{3}}{2}$ (d) $y = 2x$, $y = x + \frac{\sqrt{3}}{2} + \frac{\pi}{6}$
30. What is the x -coordinates at the point where the normal to $f(x) = x^2 - 3x + 1$ at $x = -1$ intersect the curve again? (a) 5 (b) -5 (c) $-\frac{21}{5}$ (d) $\frac{21}{5}$
31. Find the equation of the normal to the curve $y = \tan x$ at $x = \frac{\pi}{4}$
 (a) $2x - \frac{\pi}{2} + 1$ and $1 - \frac{x}{2} + \frac{\pi}{8}$ (b) $1 - \frac{x}{2} + \frac{\pi}{8}$ (c) $2x + 1 - \frac{\pi}{2}$ (d) $1 - \frac{x}{2} - \frac{\pi}{8}$ and $1 - 2x - \frac{\pi}{2}$
- Answer 32 to 35 with the following. Let the equation of the parabola be $8y^2 - x - y - 12 = 0$. Find its
32. Latus rectum (a) $\frac{1}{12}$ (b) $\frac{1}{8}$ (c) $\frac{1}{2}$ (d) $\frac{1}{4}$
33. Vertex (a) $(-\frac{1}{8}, \frac{65}{4})$ (b) $(-\frac{45}{32}, \frac{1}{4})$ (c) $(-\frac{385}{32}, \frac{1}{16})$ (d) $(16, -\frac{32}{358})$
34. Focus (a) $(-12, \frac{1}{16})$ (b) $(16, -\frac{1}{12})$ (c) $(-\frac{1}{12}, 8)$ (d) $(\frac{9}{2}, -12)$
35. Equation of the directrix (a) $x = -\frac{13}{48}$ (b) $y = -\frac{16}{193}$ (c) $x = -\frac{193}{16}$ (d) $y = -\frac{16}{49}$
36. Find the distance between the pair of points P(12, -15) and Q(3, -2). (a) $\sqrt{370}$ (b) $10\sqrt{5}$ (c) $\sqrt{221}$ (d) $5\sqrt{10}$
37. Find the slope of the line passing through the points $(\frac{1}{5}, \frac{1}{3})$ and $(\frac{1}{3}, \frac{1}{5})$. (a) 4 (b) 5 (c) -3 (d) -1
38. The intersecting angle between the lines L_1 and L_2 is 45° . If L_1 passes through the points (3, -1) and (5, 3). find the slope of L_2 . (a) -3 (b) -5 (c) 5 (d) 3
39. Find the value of p if the line joining (p, 4) and (6, -2) is perpendicular to the line joining (2, p) and (-1, -3). (a) 4 (b) 6 (c) 3 (d) 0
40. Find the value of p , if the line which passes through (-1, -p) and (-2p, 2) is parallel to the line $10y - 2x + 1 = 0$. (a) $\frac{7}{6}$ (b) $\frac{6}{7}$ (c) $-\frac{2}{7}$ (d) $-\frac{6}{7}$
41. A particle moves with a position vector $r \cos \theta i + \sin \theta j$. What is the value for which the velocity of the particle is perpendicular to ? (a) 1 (b) 0 (c) $-\theta^2 r$ (d) θk
42. Suppose particle P moves on a circle C of radius R. If s is the length of arc measures along C from A to P and θ is the corresponding angle subtended at the centre O. Find the normal acceleration of the body if v is the tangential velocity. (a) $\frac{dv}{dt}$ (b) $\frac{d^2\theta}{dt^2}$ (c) $\frac{d^2s}{dt^2}$ (d) $\frac{v^2}{R}$
43. Find the sum of the vectors: 4OA, 3AO, AB, BZ, 7OB, 7BZ. (a) 5OZ (b) 6OZ (c) 7OZ (d) 8OZ
44. Given $r_1 = 3i - 2j + k$, $r_2 = 2i - 4j + 3k$, $r_3 = -i + 2j + 2k$. Find $|2r_1 - 3r_2 - 5r_3|$ (a) $\sqrt{30}$ (b) $4\sqrt{2}$ (c) 3 (d) 5
45. Find the work done in moving an object along a vector $r = 3i + 2j + 5k$ if the applied force is $F = 2i - j - k$. (a) 8 (b) 9 (c) 7 (d) 6
46. Find the angle between the vectors $r_1 = 3i + 2j - 6k$ and $r_2 = 4i - 3j + k$ (a) 57° (b) 79° (c) 90° (d) 69°
47. Find the $r \cdot v$ if $r = 4i - 3j + k$ and $v = -2i + 4j - 3k$ (a) -23 (b) -21 (c) -20 (d) -12
48. Speed of a truck is 40 ms^{-1} , after 10 seconds its speed decreases to 20 ms^{-1} , its acceleration is (a) -1 ms^{-1} (b) -2 ms^{-1} (c) -4 ms^{-1} (d) -5 ms^{-1}
49. A car stops and then starts accelerating uniformly at rate of 3 ms^{-2} . Speed of car after 20 seconds is (a) 40 ms^{-1} (b) 30 ms^{-1} (c) 100 ms^{-1} (d) 60 ms^{-1}
50. A stone is thrown upwards with initial velocity of 20 ms^{-1} , height that stone will reach would be (a) 20 m (b) 30 m (c) 40 m (d) 50 m

Paper 4 of 4 — MTH 103 PAST QUESTIONS (VECTORS AND DYNAMIC) 2019/2020

Level 100 • • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN

COLLEGE OF SCIENCE

DEPARTMENT OF MATHEMATICS

SECOND SEMESTER EXAMINATION 2019/2020

Course code & title: MTH 123 -Vectors & Dynamics. Duration: 2HRS

15MINS

PAST QUESTIONS

- Given that $\hat{A} = 2\hat{i} + \hat{j} + 2\hat{k}$. Find the square of $|\hat{A}| + 1$. (a) 3 (b) 4 (c) 9 (d) 10
 $\sqrt{2}$ 1
- Find the unit vector for the vector $\hat{B} = 3\hat{i} - 4\hat{j} + 5\hat{k}$. (a) $\frac{1}{\sqrt{50}}(3\hat{i} - 4\hat{j} + 5\hat{k})$ (b) $\frac{1}{\sqrt{2}}(3\hat{i} - 4\hat{j} + 5\hat{k})$
 $\sqrt{2}$ 1
 $\sqrt{2}$ 1
 (c) $\frac{1}{5}(3\hat{i} + 4\hat{j} + 5\hat{k})$ (d) $\frac{1}{2}(3\hat{i} + 4\hat{j} + 5\hat{k})$
 26
- Given $\hat{A} = 4\hat{i} + \hat{j} + 3\hat{k}$ and $\hat{B} = 2\hat{j} + 8\hat{k}$. Obtain $|\hat{A} \cdot \hat{B}|$. (a) 30 (b) 26 (c) 22.8 (d) $\sqrt{520}$
 $\sqrt{2}$
- Given that $\hat{A} = 3\hat{i} - 4\hat{j} + 5\hat{k}$ and $\hat{B} = \hat{i} + 2\hat{j} + 2\hat{k}$. calculate the cosine angle between $\hat{A}$ & $\hat{B}$. (a) $\frac{6}{11}$
 1
 (b) $\frac{5}{11}$ (c) $\frac{15\sqrt{2}}{11}$ (d) $\frac{15\sqrt{2}}{11}$
- Given that $\hat{u}$ is a function for the vectors $\hat{A}$ and $\hat{B}$. Thus, if $\hat{A}(\hat{u}) \cdot \hat{B}(\hat{u}) = 0$. Then, ...
 (a) Parallel (b) unit vector (c) perpendicular (d) scalar triple product
- If $\hat{A} \cdot \hat{B} = 0$ and $\hat{A}$ and $\hat{B}$ are non-zero vectors, then $\hat{A}$ and $\hat{B}$ are said to be...
 (a) Parallel (b) unit vector (c) perpendicular (d) scalar triple product
- Which of the following represent a continuous vector
 $\hat{r}(\hat{r} + \Delta\hat{r}) - \hat{r}(\hat{r})$ $\hat{r}(\hat{r} + \Delta\hat{r}) - \hat{r}(\hat{r})$
 (a) $\lim_{\Delta\hat{r} \rightarrow 0} \Delta\hat{r}$
 (b) $\lim_{\Delta\hat{r} \rightarrow 0} \Delta\hat{r} + \hat{r}$
 (c) $\lim_{\Delta\hat{r} \rightarrow 0} \hat{r}(\hat{r}) = \hat{r}(\hat{r}0)$ (d) $\lim_{\Delta\hat{r} \rightarrow 0} \hat{r}(\hat{r}) = \hat{r}(\hat{r}0)$
 $\hat{r} \rightarrow \Delta\hat{r}$ $\hat{r} \rightarrow \Delta\hat{r}$ $\hat{r} \rightarrow 0$ $\hat{r} \rightarrow \Delta\hat{r}$
 $\hat{r} \hat{r}$ $\hat{r} \hat{r}$
- If $\hat{r}(\hat{r} + \Delta\hat{r}) = (\hat{r} * \hat{r}) + (\hat{r} * \Delta\hat{r})$. Then the differential of $\hat{A}$ is given as (a) $\hat{r} \hat{r} + \hat{r} \hat{r} + \hat{r} \hat{r}$
 $\hat{r} \hat{r} \hat{r} \hat{r} \hat{r} \hat{r} \hat{r} \hat{r} \hat{r}$
 $\hat{r} \hat{r}$ (b) $\hat{r} \hat{r} + \hat{r} \hat{r} + \hat{r} \hat{r}$ (c) $\hat{r} \hat{r} + \hat{r} \hat{r} + \hat{r} \hat{r}$ (d) $\hat{r} \hat{r} + \hat{r} \hat{r} + \hat{r} \hat{r}$
 $\hat{r} \hat{r} \hat{r} \hat{r} \hat{r} \hat{r} \hat{r} \hat{r} \hat{r}$
 $\hat{r} \hat{r} \hat{r} \hat{r} \hat{r} \hat{r} \hat{r} \hat{r}$
- Express $\hat{r} \hat{r}$ (a) as a product. (a) $\hat{r} \hat{r} + \hat{r} \hat{r}$ (b) $\hat{r} * \hat{r} + \hat{r} * \hat{r}$ (c) $\hat{r} \hat{r} + \hat{r} \hat{r}$ (d) $\hat{r} \hat{r}$
 $\hat{r} \hat{r}$

$$\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{a}$$

$$2\vec{a} \cdot 2\vec{b}$$

10. Given that $\phi(x, y, z) = xyz$. Find $\nabla\phi$. (a) $xy\vec{i} + xz\vec{j} + 2yz\vec{k}$ (b) $2xy\vec{i} + x\vec{j} + y\vec{k}$ (c)

$$2xy\vec{i} + x\vec{j} + y\vec{k} \quad (d) \quad 2xy\vec{i} + x\vec{j} + y\vec{k}$$

11. If $\phi = x^2 + y^2 + z^2 = 2x^2 + y^2 - x^2 - y^2$. Find $\nabla\phi$ at $(1, -1, 1)$. (a) $-5\vec{i} - \vec{j}$ (b)

$$5\vec{i} + \vec{j} \quad (c) \quad 2x^2 + y^2 + z^2 + x^2 - y^2 - z^2 \quad (d) \quad (-3x^2 - y^2 - z^2) - (3x^2 - y^2 - z^2)$$

$$2x^2 + y^2 + z^2$$

$$2x^2 + y^2 + z^2$$

$$(2x^2 - y^2 - z^2)\vec{i}$$

$$2x^2 + y^2 + z^2$$

12. If $\phi = x^2 + y^2 + z^2$. Evaluate $\nabla^2 \log \phi$ (a) $\frac{1}{\phi} + \frac{1}{\phi^2} + \frac{1}{\phi^3}$ (b) $\frac{1}{\phi} - \frac{1}{\phi^2} + \frac{1}{\phi^3}$ (c) $\frac{1}{\phi^2} + \frac{1}{\phi^3} + \frac{1}{\phi^4}$ (d)

$$\frac{1}{\phi} + \frac{1}{\phi^2} + \frac{1}{\phi^3}$$

$$\frac{1}{\phi}$$

$$+ \frac{1}{\phi} + \frac{1}{\phi}$$

13. Find a unit normal to the surface $x^2 + y^2 + z^2 - 4x - 4y + 4z = -7$ at $(2, 1, -2)$

$$(a) -8\vec{i} + \vec{j} \quad (b) 8\vec{i} + \vec{j} \quad (c) (2x - 4)\vec{i} + (2y - 4)\vec{j} + (2z - 4)\vec{k} \quad (d) x^2 + y^2 + z^2 - 4x - 4y + 4z$$

$$x^2 + y^2 + z^2 - 4x - 4y + 4z$$

$$x^2 + y^2 + z^2$$

14. If $\nabla\phi = 0$. ϕ is ... (a) Parallel (b) perpendicular (c) solenoidal (d) irrational

$$x^3$$

$$4x^2 + 3x + 2$$

15. If $\phi = 3x^2 + 4y^2 + 3z^2$. $\nabla \times \phi = 0$... (a) Parallel (b) perpendicular (c) solenoidal (d) irrational

$$x^2 + y^2$$

16. If $\phi = (3x^2 + 4y^2 + 5z^2) + (2x - 5y) + 4z^2$, $\nabla\phi$ at $(1, 2, 3)$ is ... (a) 38.12 (b) 48.37 (c) 50.13 (d) 27.73

$$17. \text{ If } \phi = x^2 + y^2 + z^2 - 4x - 4y + 4z, \nabla\phi \text{ at } (1, 2, 3) \text{ is ...}$$

$$x^2 + y^2$$

$$4x^2 + 3x + 2, \nabla\phi \text{ at } (1, 2, 3) \text{ is ... (a) 100.18 (b) 180.90 (c) 150.45 (d) 160.22}$$

$$\phi = 1$$

18. A particle moves in speed S at time t , its position is stated as $x = 2t + 3$, $y = t^2 + 3t$, $z = t^3 +$

$$2t^2. \text{ Find the velocity in the direction of the vector } 2\vec{i} + 3\vec{j} + 4\vec{k} \text{ when } t = 1. (a) 4\vec{i} + 3\vec{j} +$$

$$5\vec{k} \quad (b) 2\vec{i} + 5\vec{j} + 7\vec{k} \quad (c) 3\vec{i} + 5\vec{j} + 0\vec{k} \quad (d) 2\vec{i} - 3\vec{j} + 7\vec{k}$$

19. Determine the unit tangent vector for the curve $x = 3t$, $y = 2t^2$, $z = t^2 + t$ at the point $(6, 8, 6)$.

$$1\vec{i} + 1\vec{j} + 1\vec{k}$$

$$(a) (3\vec{i} - 8\vec{j} + 5\vec{k}) \quad (b) (3\vec{i} + 8\vec{j} + 5\vec{k}) \quad (c) (3\vec{i} - 8\vec{j} - 5\vec{k}) \quad (d) (-3\vec{i} - 8\vec{j} + 5\vec{k})$$

$$\sqrt{98}\vec{i} + \sqrt{98}\vec{j} + \sqrt{98}\vec{k}$$

$$2\vec{i}$$

20. If $\phi = 3x^2 + y^2 + (x + 2)z$, $\psi = 2x^2 - 3yz + (y - 2)z$, $\nabla(\phi \times \psi)$ at $(1, 2, 3)$ is ...

$$4\vec{i} + 4\vec{j} + 4\vec{k}$$

$(\square)(-14\square + 13\square - 24\square)(\square)(14\square + 13\square - 24\square)(\square)(14\square - 13\square - 24\square)(\square)(14\square + 13\square + 24\square)$

21. If the point (2, k) lies on the line with slope $m = 3$ passing through the point (1, 6), find k. (a) 12 (b).

4 (c) 9 (d) -9

22. Find the slope m of the line having the equation $y - 7 = 2(x - 3)$ and find a point on the line. (a) m

$= -2$ & (7, -3) (b) $m = 2$ & (3, 7) (c) $m = 3$ & (-2, 7) (d) $m = 7$ & (2, 3)

$\square \square \square \square \square$

23. Find an equation of the line through the points (0,2) and (3,0). (a) $3 + 2 = 1$ (b) $2 + 3 = 1$ (c) $3 +$

$1 \square \square$

$\square = 2$ (d) $3 + 2 = 0$

24. Find an equation of the line through the point (2, -3) and perpendicular to the line $4x - 5y = 7$. (a)

$2\square + 3\square = 5$ (b) $4\square + 2\square = 3$ (c) $3\square + 2\square = -4$ (d) $5\square + 4\square = -2$

25. For what value of k will the line $kx + 5y - 2k$ have slope 3? (a) 12 (b) -15 (c) -9 (d) -10

26. Write the standard equation for the circle with center (3,5) and radius 4. (a) $(\square + 5)^2 + (\square + 3)^2 =$

4 (b) $(\square - 5)^2 + (\square - 3)^2 = 16$ (c) $(\square - 3)^2 + (\square - 5)^2 = 16$ (d) $(\square - 3)^2 + (\square - 5)^2 = 4$

$3 \ 9$

27. Given $\square = \sqrt{11}$ and centre (0, -), find the equation of the circle. (a) $-2\square^2 - 2\square^2 - 18\square + 9 = 2 \ 2$

0 (b) $2\square^2 + 2\square^2 - 11\square + 9 = 0$ (c) $-2\square^2 - 2\square^2 - 11\square + 9 = 0$ (d) $2\square^2 - 2\square^2 - 18\square + 9 = 0$

28. The diameter of a circle has its coordinates as A(-6, -1) and B(16, 13). Find the centre and radius

(a) (5, 6), $r = 10$ (b) (5, 6), $r = \sqrt{55}$ (c) (6, 5), $r = 55$ (d) (-5, -6), $r = \sqrt{55}$

$2 \ 5 \ 2 \ 5$

29. Let the parabola be $4\square + 10\square^2 = 5\square + 5$. Find latus rectum (a) - 3 (b) - 8 (c) - 5 (d) - 2

$5 \ 2 \ 2 \ 5$

30. Let the parabola be $4\square + 10\square^2 = 5\square + 5$. Find length of the latus rectum. (a) (b) (c) (d)

$8 \ 3 \ 5 \ 2$

$1 \ 45 \ 45 \ 1 \ 1 \ 9 \ 1 \ 9$

31. Let the parabola be $4\square + 10\square^2 = 5\square + 5$. Find vertex . (a) (,) (b) (,) (c) (,) (d) (,)

$4 \ 32 \ 32 \ 4 \ 2 \ 16 \ 4 \ 16$

$1 \ 1 \ 45 \ 45 \ 1 \ 1 \ 1$

32. Let the parabola be $4\square + 10\square^2 = 5\square + 5$. Find focus (a) (- ,) (b) (- ,) (c) (-

$4 \ 10 \ 32 \ 32 \ 10 \ 4 \ 2$

$1 \ 9 \ 1 \ 1 \ 9$

,) (d) (4 - 10 , 16)

$10 \ 16$

2

33. Let the parabola be $4x^2 + 10x = 5y + 5$. Find equation of the directrix. (a) $160x - 241 = 0$ (b) $241x - 160 = 0$ (c) $241x - 160 = 0$ (d) $160x - 241 = 0$

34. Which of the following is not an example of projectile motion? (a) a hot air balloon drifting towards

earth. (b) a volleyball served over a net (c) a long jumper in action (d) a baseball hit by a bat.

35. Lisa goes into a jump at 12m/s and launches herself at 20% above the horizontal. How long is she

in the air before returning to earth? ($g = 9.81\text{m/s}^2$). (a) 1.5 s (b) 1.2 s (c) 0.42 s (d) 0.83 s

36. What is the path of a projectile? (a) a hyperbola (b) projectiles do not follow a predictable path (c)

a wavy line (d) a parabola

37. A body of mass 10 kg, projected at an angle of 30° from the ground with an initial velocity of 5m/s,

acceleration due to gravity is $g = 10\text{m/s}^2$, what is the time of flight? (a) 0.866 s (b) 1.86 s (c) 1.96 s (d) 1.862 s

38. When do we get maximum height in a simple projectile motion? (a) When $\theta = 45^\circ$ (b) When $\theta = 60^\circ$ (c) When $\theta = 90^\circ$ (d) When $\theta = 0^\circ$

39. A soccer ball is projected at an angle of 60° from the ground. It attains its maximum height in 10s.

Considering acceleration due to gravity as $g = 10\text{m/s}^2$. What is the velocity with which it was projected? (a) 11.55 m/s (b) 11.5 m/s (c) 1.155 m/s (d) 12.0 m/s

40. An elastic string of natural length 5m and modulus 10N is cut into parts of length 2m and 3m. Each

part is tied to the same particle of mass m. Their free ends are tied to two fixed points a distance 7m apart on a smooth horizontal table. Find the period of oscillation when the particle is displaced

5m (a) $2\pi\sqrt{\frac{3}{g}}$ (b) $2\pi\sqrt{\frac{6}{g}}$ (c) $2\pi\sqrt{\frac{7}{g}}$ (d) $2\pi\sqrt{\frac{5}{g}}$

along the string. (a) $2\pi\sqrt{\frac{3}{g}}$ (b) $2\pi\sqrt{\frac{6}{g}}$ (c) $2\pi\sqrt{\frac{7}{g}}$ (d) $2\pi\sqrt{\frac{5}{g}}$

41. Two spheres of masses 3 kg and 4 kg on a smooth horizontal table are connected with a light inextensible string of length a. Initially the string is slack and the 4 kg mass is projected along the

table with velocity 21 m/s. Find the velocity with which 3 kg mass begins to move. (a) 7 m/s (b) 12 m/s (c) 17 m/s (d) 5 m/s

42. Two spheres of masses 3 kg and 4 kg on a smooth horizontal table are connected with a light inextensible string of length a. Initially the string is slack and the 4 kg mass is projected along the table with velocity 21 m/s. Find the impulse generated in the string when it suddenly becomes taut.

(a) 36 Newton-sec (b) 12 Newton-sec (c) 6 Newton-sec (d) 10 Newton-sec

43. Find the length of a simple pendulum if its period for small oscillations is 1 sec. Take acceleration

due to gravity as 10m/s^2 . (a) 0.253 m (b) 0.005 m (c) 25 m (d) none of the above

44. If the length of a simple pendulum is increased by 1% by what percentage is its period for small oscillation increased. (a) 1 % (b) 0.4988 % (c) 50% (d) none of the above

45. The length of a simple pendulum is 0.25 m and its maximum angle of swing is 8° . What is the

frequency of oscillation? (a) 7 Hz (b) 2.008 Hz (c) 3.667 Hz (d) 1.007 Hz

46. A body of mass 4 kg is hung on a light spring and is found to stretch it 3 cm. The mass is the pulled

down a further 2 cm and released. Find the period of oscillation of the mass. (a) 0.3444 sec (b) 0.333 sec (c) 0.4555 sec (d) 7 sec

47. A body of mass 4 kg is hung on a light spring and is found to stretch it 3 cm. The mass is the pulled

down a further 2 cm and released. Find its maximum speed. (a) 0.5 m/s (b) 0.72 m/s (c) 0.365 m/s (d) 1.5 m/s

48. An elastic spring extends a distance Δ when a given mass is attached to its lowest point. The mass

is pulled down a further distance Δ ($< \Delta$) and let go. Find the period of the simple harmonic motion Δ

that ensues. (a) $\pi\sqrt{\Delta}$ (b) $2\pi\sqrt{\Delta}$ (c) $2\pi\sqrt{\Delta}$ (d) none of the above

49. A truck with a momentum of 85000 kgm/s is moving with a velocity of 50 m/s. Determine its mass

(a) 1700 kg (b) 850 kg (c) 70 kg (d) 180 kg

50. What force will give a mass of 8kg a velocity of 15m/s in 1 minute starting from rest. (a) 2N (b) 5N (c) 7.5N (d) 12N

51. Find the acceleration produced when a force of 5N acts on a mass of 10kg (a) 12m/s (b) 7m/s (c) 0.5m/s

(d) 3m/s

52. A particle moves along a curve C such that its position vector is given by $\mathbf{r} = 5\mathbf{i} + 3t^2\mathbf{j} + 5t^3\mathbf{k}$.

Find its acceleration at the anytime t. (a) $6\mathbf{i} + 30t\mathbf{j}$ (b) $5\mathbf{i} + 6t\mathbf{j} + 15t^2\mathbf{k}$ (c) $5\mathbf{i} + 3t^2\mathbf{j} + 5t^3\mathbf{k}$

(d) $6\mathbf{j}$

53. Given a space curve C with position vector $\mathbf{r} = 2t\mathbf{i} + 6t\mathbf{j} + 9t\mathbf{k}$. Find the radius of curvature (a)

1/25

11 (b) 11 (c) 3 (d) 10

54. Find the unit principal normal of $\mathbf{r} = 2t\mathbf{i} + 6t\mathbf{j} + 9t\mathbf{k}$.

1

(a) $11(2t + 6t + 9t)$ (b) $11(2t + 6t + 9t)$ (c) $2t + 6t + 9t$ (d) $-\cos 2t - \sin 2t$

55. A particle moves with a position vector $\mathbf{r} = \cos t\mathbf{i} + \sin t\mathbf{j}$. What is the value for which the velocity

of the particle is perpendicular to ? (a) 1 (b) 0 (c) $2r$ (d) $\mathbf{k}$

56. Suppose particle P moves on a circle C of radius R. If s is the length of arc measures along C from

A to P and θ is the corresponding angle subtended at the centre O. Find the normal acceleration of $\frac{ds}{dt}^2 - \frac{v^2}{R^2}$

the body if v is the tangential velocity. (a) $\frac{v^2}{R}$ (b) $\frac{v^2}{R^2}$ (c) $\frac{v^2}{R^2}$ (d) $\frac{v^2}{R}$

57. Find the sum of the vectors: 4OA, 3AO, AB, BZ, 7OB, 7BZ. (a) 5OZ (b) 6OZ (c) 7OZ (d) 8OZ

58. Given $\vec{r}_1 = 3\hat{i} - 2\hat{j} + \hat{k}$, $\vec{r}_2 = 2\hat{i} - 4\hat{j} - 3\hat{k}$, $\vec{r}_3 = -\hat{i} + 2\hat{j} + 2\hat{k}$. Find $|2\vec{r}_1 - 3\vec{r}_2 - 5\vec{r}_3|$ (a) $\sqrt{30}$ (b) $4\sqrt{2}$ (c) 3 (d) 5
59. Find the work done in moving an object along a vector $\vec{r} = 3\hat{i} + 2\hat{j} - 5\hat{k}$ if the applied force is $\vec{F} = 2\hat{i} - \hat{j} - \hat{k}$ (a) 8 (b) 9 (c) 7 (d) 6
60. Find the angle between the vectors $\vec{r}_1 = 3\hat{i} + 2\hat{j} - 6\hat{k}$ and $\vec{r}_2 = 4\hat{i} - 3\hat{j} + \hat{k}$ (a) 57° (b) 79° (c) 90° (d) 69°
61. Find the $\cdot$ if $\vec{r} = 4\hat{i} - 3\hat{j} + \hat{k}$ and $\vec{s} = -2\hat{i} + 4\hat{j} - 3\hat{k}$ (a) -23 (b) -21 (c) -20 (d) -12
62. Speed of a truck is 40 ms^{-1} , after 10 seconds its speed decreases to 20 ms^{-1} , its acceleration is (a) -1 ms^{-2} (b) -2 ms^{-2} (c) -4 ms^{-2} (d) -5 ms^{-2}
63. A car stops and then starts accelerating uniformly at rate of 3 ms^{-2} . Speed of car after 20 seconds is (a) 40 ms^{-2} (b) 30 ms^{-2} (c) 100 ms^{-2} (d) 60 ms^{-2}
64. A stone is thrown upwards with initial velocity of 20 ms^{-1} , height that stone will reach would be (a) 20 m (b) 30 m (c) 40 m (d) 50 m
65. If initial velocity of an object is zero, then distance covered by it in time t and acceleration of 9.8 ms^{-2} is (a) $2.9 t^2$ (b) $3 t^2$ (c) $4 t^2$ (d) $4.9 t^2$
66. Horizontal distance travelled by a ball if it's thrown with initial velocity of 20 ms^{-1} at an angle of 30° is (a) 24 m (b) 56 m (c) 35.3 m (d) 36.3 m
67. Find the midpoint of the line segment joining the points $(4, -2)$ and $(-8, 6)$ (a) (6,4) (b) (-6,4) (c) (2,2) (d) (-2,2)
68. Find the distance between the points $(-3, 2)$ and $(6, 4)$ (a) $\sqrt{85}$ (b) $\sqrt{79}$ (c) $\sqrt{3}$ (d) $\sqrt{5}$
69. What is the gradient of the line passing through the points (4,6) and (-1,-2)? (a) 3 (b) 4 (c) 5 (d) 8
70. Which points satisfies the linear quadratic system $y = x + 3$ and $y = 5 - x^2$? (a) (-2,1) (b) (2,1) (c) (-1,2) (d) (4,-1)